

WEB TECHNOLOGIES

CSE-4004

LECTURE 1: Web Storage, Introduction to Canva

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WEB STORAGE

- Used to store data inside the browser without relying on server.
- 1. **localStorage**: Used to save login related information. It is stored permanently until deleted. *Eg.: Theme, language preferences, username etc.*
- 2. **sessionStorage**: Used to save session related information. Data is deleted when tab closes. *Ex: loginstatus, sessionID*

We can access these built in variables of a browser by writing JavaScript code.

Example:

```
<script>  
    localStorage.setItem("username", "VIT-AP");  
    document.write(localStorage.getItem("username"));  
</script>
```

WEB STORAGE

Example:

```
<script>
```

```
    sessionStorage.setItem("username", "siraj");
```

```
    document.write(sessionStorage.getItem("username"));
```

```
</script>
```

```
<html>  
  <head>  
    <title>Web Storage</title>  
  </head>  
  <body>  
    <script>  
      localStorage.setItem("Institute","VIT-AP");  
      alert("Data Saved in Local Storage");  
      sessionStorage.setItem("username","Siraj");  
      alert("Data stored in Session Storage");  
      document.write("<h2>" + localStorage.getItem("Institute") + "<br>");  
      document.write("<h2>" + sessionStorage.getItem("username"));  
    </script>  
  </body>  
</html>
```

CANVAS

- HTML Canvas element is used to draw shapes and create animations on a webpage using JavaScript.

Example:

```
<canvas id="myCanvas" width="400" height="300"  
style="border:1px solid black;"></canvas>
```

```
<script>
```

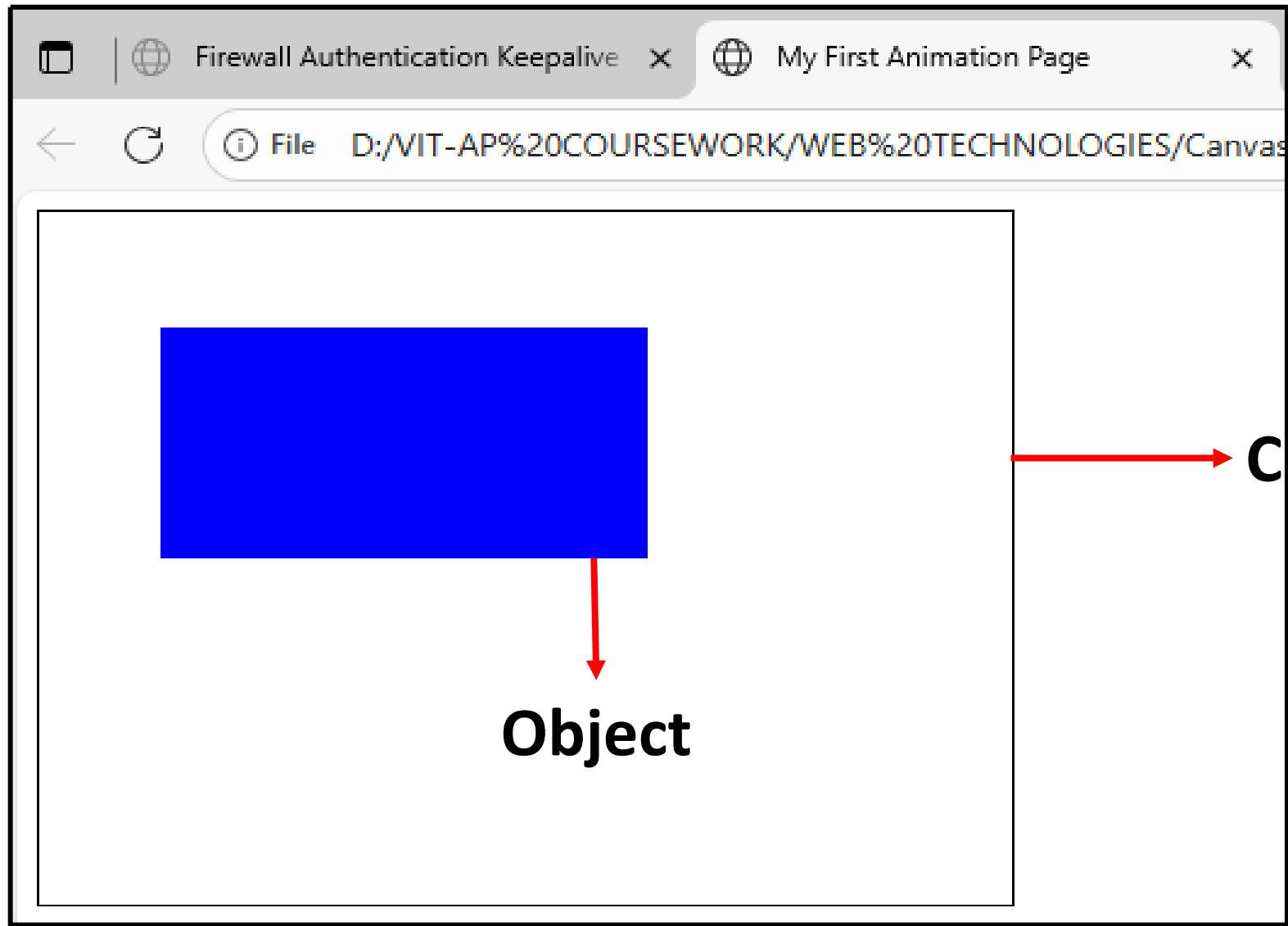
```
let c = document.getElementById("myCanvas");
```

```
let ctx = c.getContext("2d");
```

```
ctx.fillStyle = "blue";
```

```
ctx.fillRect(50, 50, 200, 100);
```

```
</script>
```



AUDIO

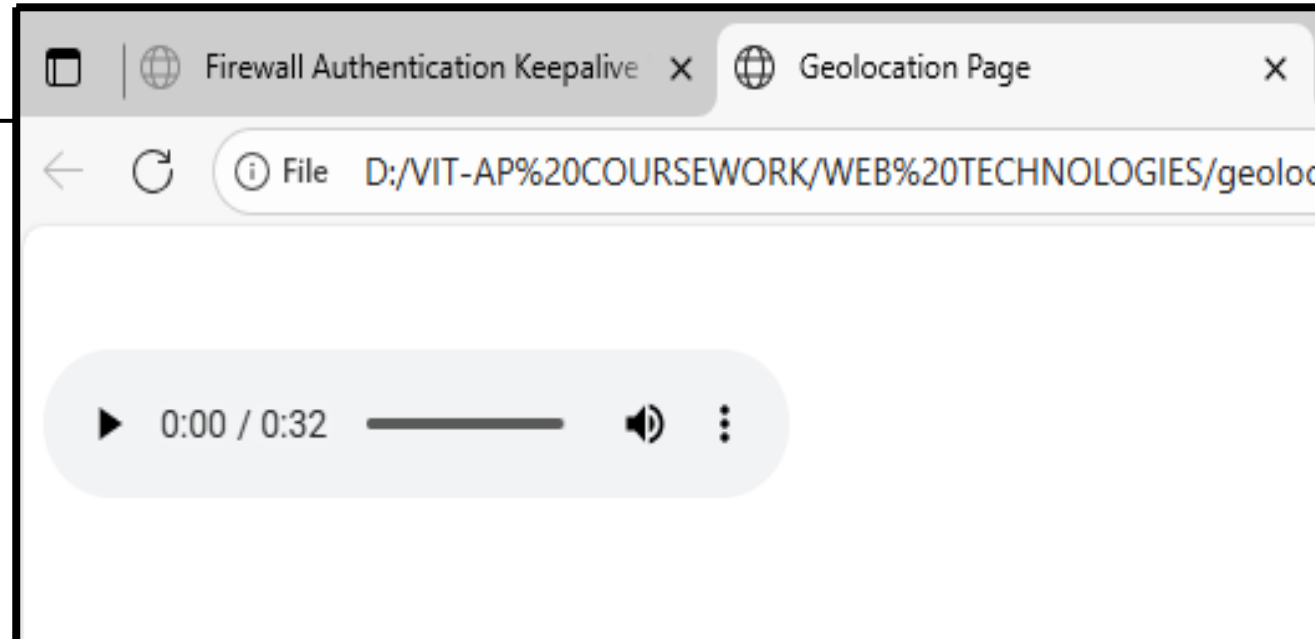
- The HTML5 <audio> element is used to play audio files in a webpage without any plugins.
- It also provides built-in controls.
- It supports MP3, WAV & OGG formats.

Example:

<audio controls>

<source src="music.mp3" type="audio/mpeg">

</audio>



VIDEO

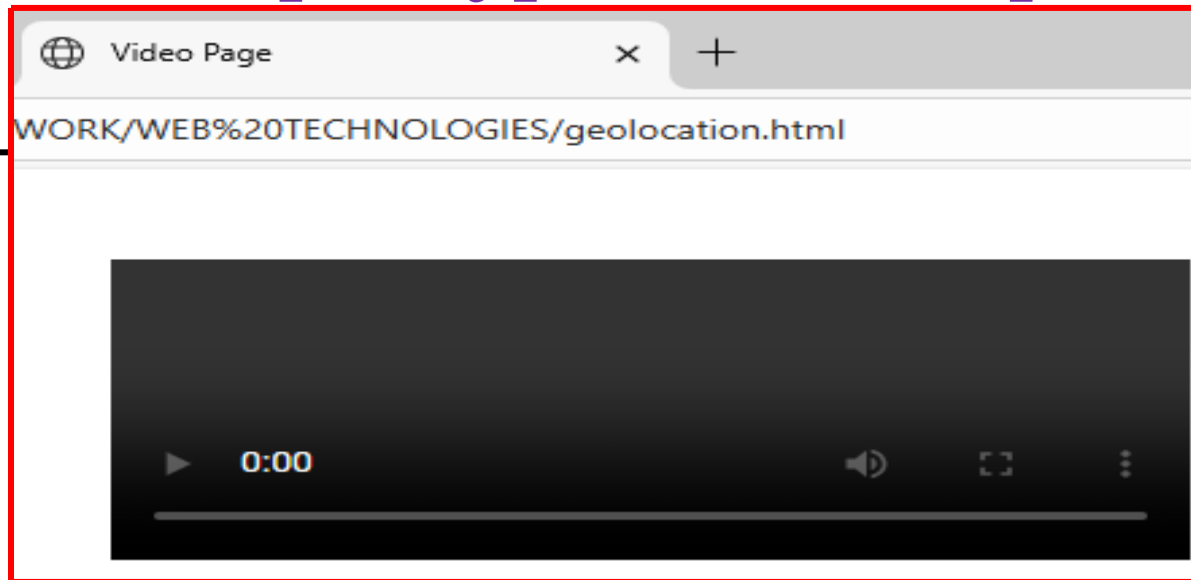
- The HTML5 <video> element is used to play video files in a webpage without any plugins.
- It also provides built-in controls.
- It supports MP4, WebM & OGG formats.

Example:

<video width="400" controls>

<source src="movie.mp4" type="video/mp4">

</video>



GEOLOCATION API

- The Geolocation API is an HTML5 feature that allows a webpage to get the location of the user.
- It needs user permission.

```
<button onclick="getLocation()">Get My Location</button>
<p id="demo"></p>
<script>
function getLocation() {
    navigator.geolocation.getCurrentPosition(show);
}
function show(position) {
    document.getElementById("demo").innerHTML =
        "Latitude: " + position.coords.latitude +
        "<br>Longitude: " + position.coords.longitude;
}
</script>
```

DRAG & DROP

- *HTML5 provides a way to drag an element and drop it into another part of the webpage.*

STEPS

1. Select a Drop Area (target)
2. Make an element draggable
3. Store ID of the dragged element
4. Drop dragged element inside the target area.

Note: Browsers do not allow dropping by default.
`event.preventDefault()`

DRAG & DROP

- *HTML5 provides a way to drag an element and drop it into another part of the webpage.*

```
<div id="dropBox"
style="width:300px; height:200px; border:2px dashed black;"
ondragover="event.preventDefault()"
ondrop="drop(event)">
</div>
```

```

```

DRAG & DROP

```
<script>  
function drop(event)  
{  
  event.preventDefault();  
  let id = event.dataTransfer.getData("text");  
  event.target.appendChild(document.getElementById(id));  
}  
</script>
```

DRAG & DROP

